

YASAWA-I-RARA, Fiji

WMO: 916600

Lat: 16.700S

Lon: 177.583E

Elev: 50

StdP: 100.73

Time Zone: 12.00 (PCF)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.5	22.1	12.0	8.8	25.0	15.2	10.9	24.1	14.0	23.9	13.1	24.4	6.4	130	0.500

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.0	33.2	27.4	32.4	27.2	31.6	27.0	30.3	30.9	29.3	30.6	28.6	30.2	3.8	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.2	27.8	30.6	29.1	25.9	29.9	28.2	24.6	29.4	102.3	30.7	96.8	30.6	93.4	30.5	33.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.4	11.4	10.6	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.0	28.3	28.7	28.2	27.6	26.9	26.2	25.4	25.3	25.9	26.3	27.1	27.9		
	DBStd	1.61	1.29	1.33	1.04	1.01	1.09	1.21	1.12	1.33	1.14	1.11	1.08	1.06		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	6199	568	523	566	528	525	486	478	474	477	505	513	554		
	CDD18.3	3157	310	289	307	278	267	236	220	216	227	247	263	296		
	CDH23.3	28632	3492	3336	3393	2839	2326	1756	1284	1257	1512	1889	2429	3122		
	CDH26.7	7107	1161	1195	1067	693	434	256	147	153	244	333	538	886		
(14) Wind	WSAvg	5.5	5.1	4.6	4.4	5.5	5.3	5.5	6.1	6.5	5.7	5.9	5.4	5.8		
(15) Precipitation	PrecAvg	2063	308	269	280	214	144	103	66	74	93	136	158	217		
	PrecMax	2732	841	479	489	465	341	290	154	204	232	281	401	408		
	PrecMin	1303	73	36	84	30	16	17	4	7	17	16	50	41		
	PrecStd	379	161	101	101	114	79	71	35	47	51	70	87	81		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.2	35.5	33.2	32.4	32.1	31.2	30.7	30.2	31.1	31.4	32.3	33.0		
		MCWB	28.1	27.9	27.9	26.7	27.0	25.7	24.6	25.4	25.6	25.7	25.8	26.2		
	2%	DB	33.1	33.5	32.2	31.3	30.9	30.0	29.2	29.1	29.8	30.1	31.0	31.7		
		MCWB	28.1	27.2	28.0	27.1	26.3	25.2	24.5	24.8	25.5	25.7	25.5	26.2		
	5%	DB	31.9	32.4	31.4	30.5	29.8	29.1	28.3	28.3	29.0	29.3	30.2	31.0		
		MCWB	27.3	27.3	27.8	26.9	25.9	25.0	24.5	24.4	24.9	25.2	25.6	26.4		
	10%	DB	30.9	31.3	30.7	29.7	28.9	28.2	27.4	27.5	28.1	28.5	29.4	30.2		
		MCWB	27.1	27.2	27.5	26.6	25.6	24.7	24.2	23.9	24.3	24.7	25.6	26.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.9	31.3	31.4	30.2	30.4	28.1	27.7	27.7	28.6	28.3	29.0	30.3		
		MCDB	32.4	31.7	31.4	30.4	30.6	28.8	28.2	28.4	28.9	29.0	29.9	30.6		
	2%	WB	30.0	29.4	29.8	28.7	28.5	27.1	26.7	26.4	26.7	27.1	27.9	28.9		
		MCDB	30.7	31.6	30.7	29.6	28.9	28.0	27.3	27.7	28.3	28.8	29.1	29.8		
	5%	WB	28.7	28.5	28.8	27.9	27.2	26.3	26.0	25.6	25.8	26.2	27.2	28.1		
		MCDB	30.7	30.9	30.5	29.0	28.1	27.4	26.8	27.0	27.7	27.9	28.4	29.6		
	10%	WB	28.0	27.9	28.1	27.3	26.5	25.7	25.3	24.9	25.1	25.7	26.6	27.4		
		MCDB	29.9	29.9	29.8	28.6	27.7	26.9	26.1	26.2	26.8	27.2	27.9	28.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.0	5.0	5.0	4.3	4.3	4.0	4.1	4.1	4.6	4.5	4.5	4.6		
		MCDBR	6.9	6.8	6.0	5.4	5.8	5.3	5.5	5.3	5.6	5.6	5.6	5.7		
		MCWBR	3.5	3.5	3.5	3.3	3.2	2.6	2.7	3.0	2.8	2.7	2.6	2.9		
	5% WB	MCDBR	6.0	5.8	5.5	4.5	4.7	3.9	3.9	4.6	4.9	4.7	4.4	4.9		
		MCWBR	4.0	3.6	4.1	3.7	3.9	2.7	2.9	3.6	3.2	2.7	3.0	3.4		
(40) Clear-Sky Solar Irradiance	taub	0.387	0.384	0.375	0.375	0.372	0.371	0.366	0.366	0.378	0.382	0.382	0.382	0.382		
	taud	2.632	2.648	2.689	2.677	2.653	2.637	2.640	2.643	2.602	2.611	2.632	2.645	2.645		
	Ebn at Noon	957	951	938	904	871	854	869	899	921	941	956	962	962		
	Edh at Noon	101	99	92	88	84	83	84	89	99	101	101	101	100		
(44) All-Sky Solar Radiation	RadAvg	6.29	5.95	5.55	4.94	4.40	4.12	4.36	4.89	5.58	6.06	6.26	6.36	6.36		
	RadStd	0.67	0.54	0.46	0.38	0.37	0.26	0.22	0.43	0.36	0.41	0.62	0.32	0.32		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	+0.18	N/A	N/A	N/A	N/A	+0.20	+0.24	N/A	N/A	+67

Nomenclature: See separate page